



MASTG

MASVS-PLATFORM

MASVS-CODE

MASVS-RESILIENCE

MASVS-PRIVACY

MASTG-TEST-0206: Undeclared PII in Network Traffic Capture

MASTG-TEST-0254: Dangerous App Permissions

MASTG-TEST-0255: Permission Requests Not Minimized

MASTG-TEST-0256: Missing Permission Rationale

MASTG-TEST-0257: Not Resetting Unused

P ANDROID MASWE-0117 TEST

MASTG-TEST-0254: Dangerous App Permissions

Overview

In Android apps, permissions are acquired through different methods to access information and system functionalities, including the camera, location, or storage. The necessary permissions are specified in the `AndroidManifest.xml` file with `<uses-permission>` tags.

Steps

There are multiple tools that can help in finding permissions in use by an app. Refer [Obtaining Compiler-Provided Security Features](#) to and use any of the mentioned tools.

1. Extract the `AndroidManifest.xml` file from the APK (see [Obtaining Information from the AndroidManifest](#)).
2. Obtain the list of declared permissions (see [Obtaining App Permissions](#)).

Observation

The output should contain the list of permissions declared by the app.

Evaluation

The test fails if there are any dangerous permissions in the app.

Compare the list of declared permissions with the list of [dangerous permissions](#) defined by Android. You can find more details in the [Android documentation](#).

Context Consideration:

Context is essential when evaluating permissions. For example, an app that uses the camera to scan QR codes should have the `CAMERA` permission. However, if the app does not have a camera feature, the permission is unnecessary and should be removed.

Also, consider if there are any privacy-preserving alternatives to the permissions used by the app. For example, instead of using the `CAMERA` permission, the app could [use the device's built-in camera app](#) to capture photos or videos by invoking the `ACTION_IMAGE_CAPTURE` or `ACTION_VIDEO_CAPTURE` intent actions. This approach allows the app to access the camera functionality without directly requesting the `CAMERA` permission, thereby enhancing user privacy.

Demos

MASTG-DEMO-0033: Dangerous Permissions in the AndroidManifest with semgrep